

Isogeny volcano of a Koblitz curve: Pollard rho on equal terms ($m = 41$)

Report dated 2026-10-06. Study directory `research/isogeny_volcano_rho_m41_20261006/`; this replicates `research/isogeny_volcano_rho_20261005/` ($m = 37$). It is a standalone study: no H-*, EV-*, DEC-* or RUN-* record was written, and it is not a sealed `measured_bound`. The run artifacts cited below are the files in the study directory.

Status labels: **DERIVED** follows by proof or exact computation from stated facts; **MEASURED** observed in this study's runs; **INDEPENDENTLY VERIFIED** checked by a second, independent method; **PROPOSED** a hypothesis, not shown here.

1. Question and search scope

Is any curve isogenous to the Koblitz curve $E_1 : y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{41}}$, at any level of its isogeny volcano, better for Pollard rho when compared on equal terms (same group, solver and cost accounting)? Search scope: the whole $a_2 = 0$ isogeny class (708,153 curves), every level, all endomorphisms whose degree is at most 2^{44} and factors over primes ≤ 43 , and the GHS descent invariant.

2. Object (exact identities)

- Field $\mathbb{F}_2 \frac{[x]}{x^{41}+x^3+1}$; a curve $E_b : y^2 + xy = x^3 + b$ is named by the integer of its bits. **DERIVED**
- $\#E = 4 \cdot \ell$, $\ell = 549756390943$ (prime, $\approx 2^{39}$), trace -2308219 . **DERIVED**
- $[\mathcal{O}_K : \mathbb{Z}[\pi]] = 409 \cdot 1721$, $K = \mathbb{Q}(\sqrt{-7})$, $h(\mathcal{O}_K) = 1$, both primes inert. **DERIVED**
- Levels: crater 1 curve; 409-floor 410; 1721-floor 1,722; bottom 706,020. **DERIVED**
- Embedding degree 6,704,346,231; twist order $2 \cdot 739 \cdot 2543 \cdot 585071$; cyclic group; $\text{Aut} = \{\pm 1\}$. **DERIVED**

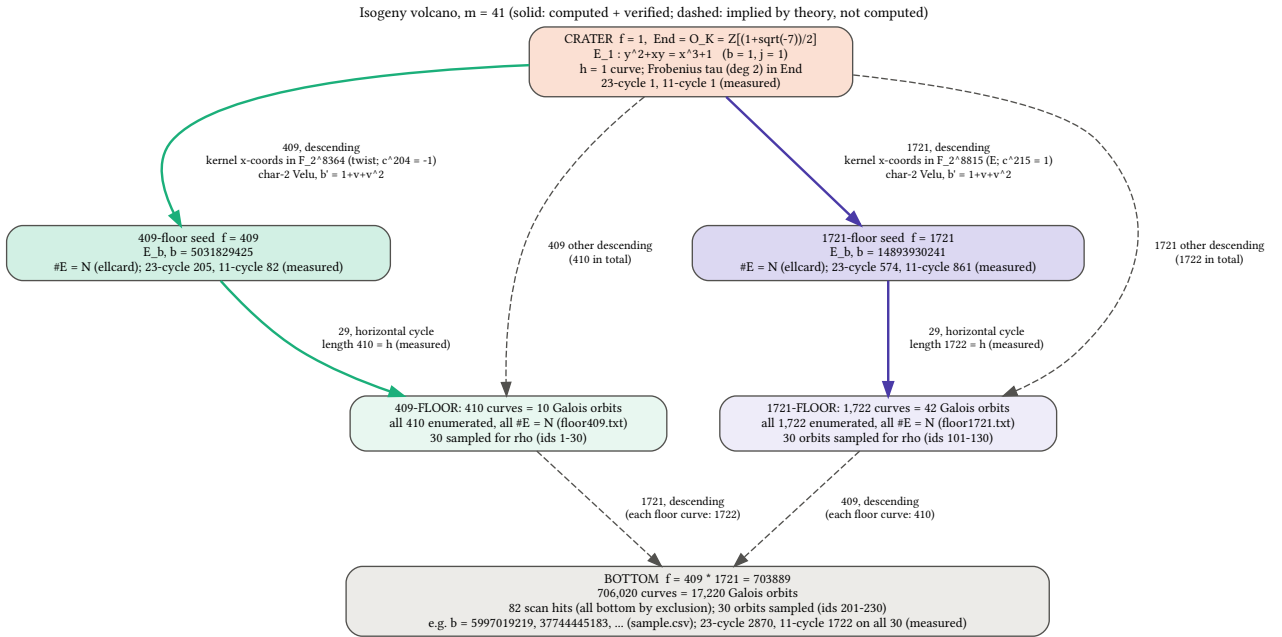


Figure 1: Volcano and the isogenies used. Solid: computed here, with endpoint verified ($\#E = N$, both cycle invariants). Dashed: implied by volcano theory, not computed. Source: `volcano_diagram.dot`.

3. Methods

- Vertical isogenies** (`vert_lib.gp`, `v409.gp`, `v1721.gp`). Kernel generators were found by x-only López–Dahab ladders in $\mathbb{F}_{2^{8364}}$ (409, quadratic twist) and $\mathbb{F}_{2^{8815}}$ (1721, E). The image comes from char-2 Velu: $b' = \frac{1}{j} = 1 + v + v^2$, with v the sum of the kernel x-coordinates. **DERIVED** The formula was checked against PARI `ellisogeny` on 6 random 73-isogenies at $m = 37$: 6/6 exact. **INDEPENDENTLY VERIFIED** Results: 409-floor $b = 5031829425$ and 1721-floor $b = 14893930241$, both with $\#E = N$ by `ellcard`. **MEASURED**
- Floor enumeration** (`floors.gp`). One horizontal 29-isogeny cycle from each seed: lengths 410 and 1722, equal to h , giving all 410 and all 1,722 floor curves, each with $\#E = N$. **MEASURED**
- Bottom** (`search.c`, `select.py`). A scan of $3 \cdot 10^8$ random b gave 82 class members, all on the bottom by exclusion from the fully enumerated floors. **MEASURED**

4. **Level check on all 91 sampled curves** (chk_*.out). $\#E = N$, plus the measured 23- and 11-cycle lengths: 1/1, 205/82, 574/861, 2870/1722. 91/91 match the class-group orders. **INDEPENDENTLY VERIFIED**
5. **Rho** (rho.c, params.h). van Oorschot-Wiener distinguished points, $\theta = 2^{-5}$, a 128-adding walk with look-ahead and cycle escape; every group operation after Q is known is counted; every solve is verified against the secret and by recomputing $[k]P = Q$. Samples: 3,000 solves of the crater in each mode, and 200 on each of 30 curves per lower level. **MEASURED**

4. Results

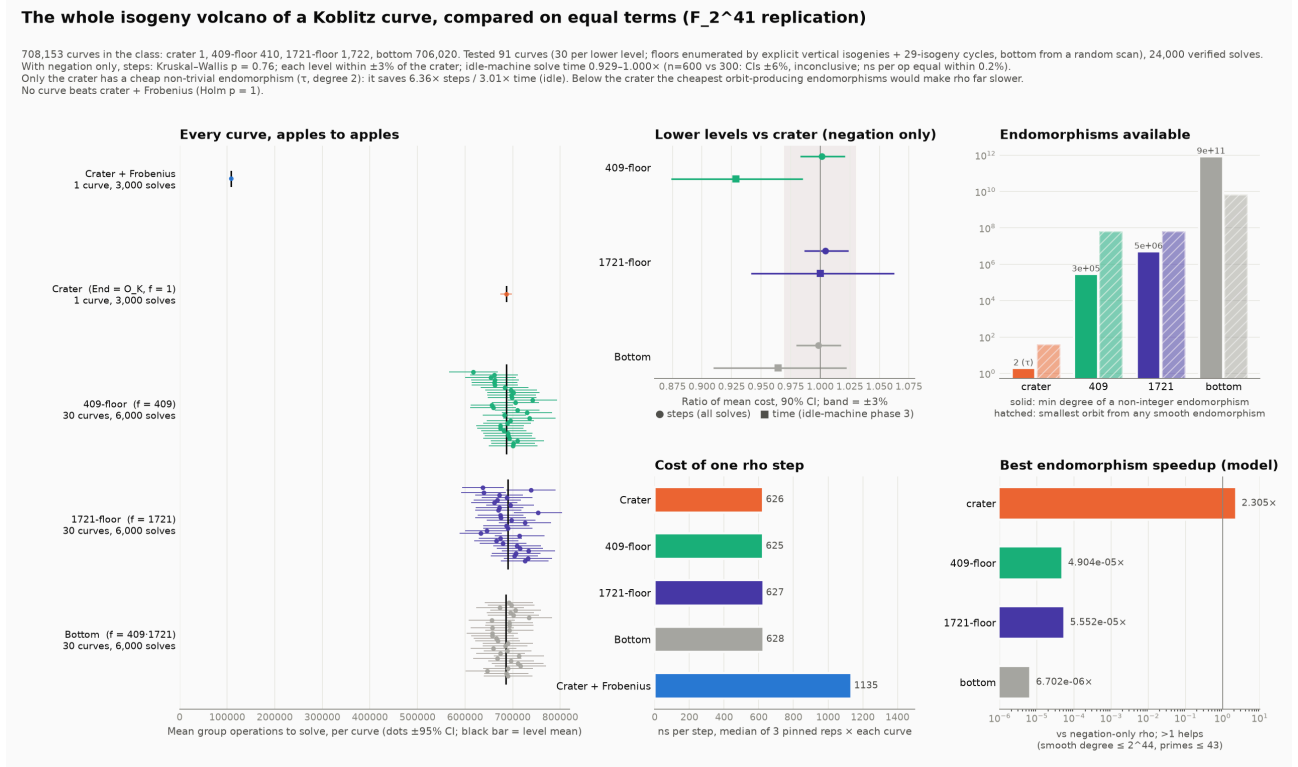


Figure 2: Mean group operations per curve (95% CI), level ratios against the crater (90% CI, $\pm 3\%$ band), cost per step, available endomorphisms, and the modelled best class speedup. Data: results.json, data/raw_solves.tar.gz, data/raw_idle.tar.gz; sample sizes in panels.

Positive or negative? Negative for the question as posed: no curve below the crater is better.

- All 24,000 solves verified. **MEASURED**
- **Negation only, steps.** No level effect (Kruskal-Wallis $p = 0.76$; ANOVA on curve means $p = 0.83$). Ratios against the crater, with 90% CIs: 409-floor 1.001 [0.983, 1.021], 1721-floor 1.005 [0.986, 1.024], bottom 0.998 [0.980, 1.017]. All equivalent within $\pm 3\%$ by TOST. **MEASURED**
- **Main-run time:** also equivalent within $\pm 3\%$ (0.997, 1.000, 0.992; $p = 0.52$). **Cost per step:** within 0.4% ($p = 0.50$). **MEASURED**
- **Frobenius on the crater:** 6.36 $\times$ fewer steps [6.20, 6.53], against $\sqrt{41} = 6.40$, and 3.0 $\times$ less time. No curve is faster than crater + Frobenius (Holm-adjusted $p = 1.0$, all 91 curves). **MEASURED**
- **Endomorphisms.** $\tau^k \in \text{End}$ only for 41 | k below the crater. **DERIVED** The minimal non-integer endomorphism degree is 2 (crater), 292,742 (409-floor), 5,183,222 (1721-floor) and $8.7 \cdot 10^{11}$ (bottom). **DERIVED** The smallest orbit from any smooth endomorphism is 41 (crater, τ), then $6.8 \cdot 10^7$, $6.8 \cdot 10^7$ and $6.7 \cdot 10^9$. **DERIVED** With the stated cost model, the best class speedup is 2.31 $\times$ (crater) and $\leq 6 \cdot 10^{-5} \times$ below it. That cost model is **PROPOSED**, not measured.
- **GHS:** $m(b) = 1$ at the crater and 41 on all 90 sampled lower curves. **DERIVED**

5. Contradictions and limitations

- **Phase 3 (idle-machine timing) is inconclusive.** Its time ratios were 0.929 [0.874, 0.985], 1.000 [0.941, 1.063] and 0.964 [0.910, 1.022]; the CIs exceed $\pm 3\%$. Splitting the time into factors, cost per operation is equal within 0.2%, and

the 409-floor deviation is step-count noise: its 600 solves needed 7.3% fewer steps, while 6,000 main-run solves give 1.001. **MEASURED** That phase was underpowered, at 300–600 solves per arm.

- At $m = 37$ a load confound caused a timing failure, which an amendment then resolved. It did not recur here: nothing else ran during the timed phases. **MEASURED**
- **Pre-data bugs, all caught by built-in checks:** a `Q2[]` table overflow in `rho.c` (the verifier rejected the solves); a Frobenius-power error and a root-finding type error in `vert.gp`. See `PRE_DATA_FIXES.md`, which also corrects an overstated sentence in the frozen `PROTOCOL.md`.
- **Scale:** $m = 41$, this solver, this machine. Transfer to ECC2K-130 ($m = 131$, conductor $263 \cdot 146505763881528721$) rests on exact facts, not on measurement: the class shares ℓ ; $\text{Aut} = \{\pm 1\}$ throughout; $h(\mathcal{O}_K) = 1$ places τ only at the crater; and non-integer endomorphisms in $\mathbb{Z} + f\mathcal{O}_K$ have degree $\geq 7f^2/4$. **DERIVED**

6. Next checks

- Rerun phase 3 with about 3,000 solves per arm, so the time intervals reach $\pm 3\%$. **PROPOSED**
- Re-express this result as a sealed `measured_bound` (constant with an interval) through the harness in `aburan28/crypto`, if a Coordinator wants it in the evidence record. **PROPOSED**
- Carry the GHS and endomorphism-orbit computations to the ECC2K-130 class itself; they are exact and need no `rho` runs. **PROPOSED**